Notes on a Decade of Remote Sensing Data for Fires in Yosemite National Park – DRAFT

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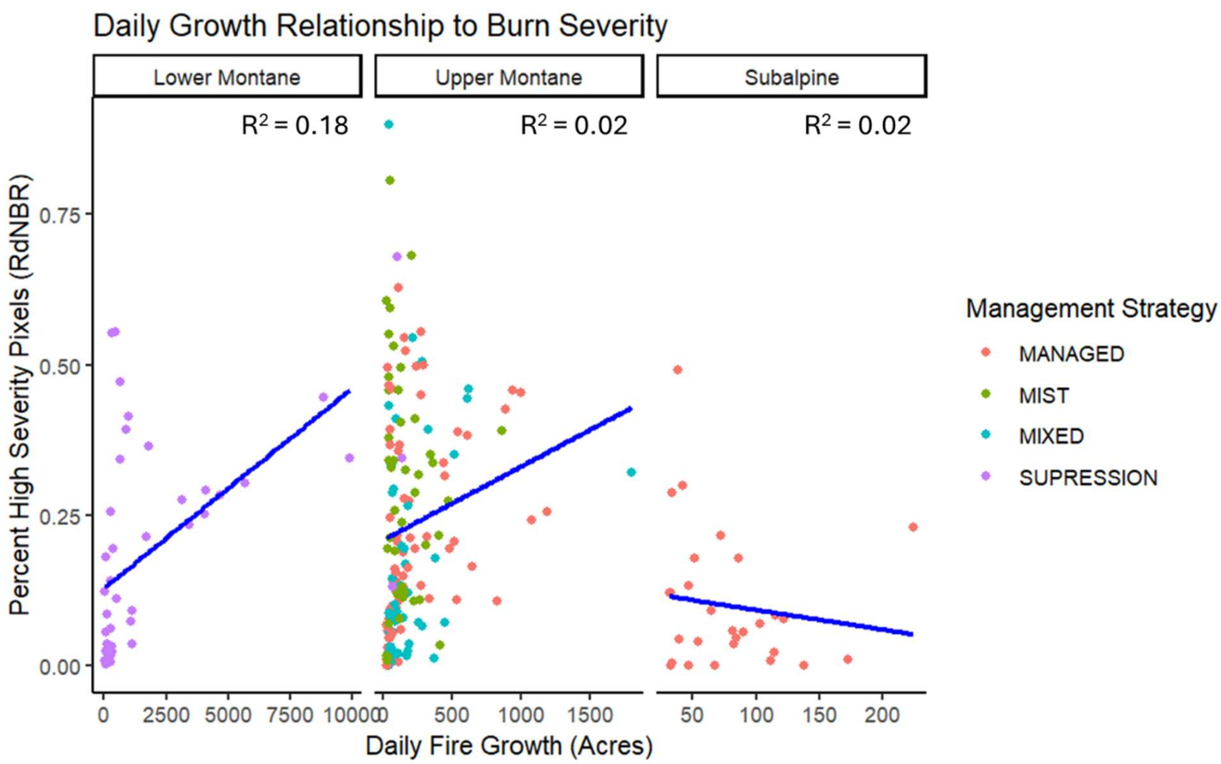
ALL ANALYSES ARE PRELIMINARY AND INTENDED AS A TRAINING EXERCISE



Introduction

The Yosemite Fire Management Plan calls for the park to burn ~16,000 acres per year if it is to re-establish the pre-colonization fire regimes that structured the park's original vegetation cover (YNP Fire Effects Monitoring Plan, 2024). These targets include estimates of the proportion that should burn at high severity, and the precise proportions and fire return intervals depend heavily on the vegetation type (Table 1). In the past nine years (2015-2023), on-park fires have burned a total of 92,583 acres, 29.4%[1] of which burned at high severity overall (Miller and Thode 2007), Table 2). This trend is largely consistent with a recent study of the Sierra Nevada which concluded that while the overall burned acreage is approaching the pre-colonization fire regime, the proportion of high severity fire is significantly higher (Williams, et al. 2023).

Yosemite fire management policy currently has three main prongs:

1) High-elevation wilderness tactics that actively manage lightning-ignited fires for resource benefit or with minimal impact suppression techniques to preserve wilderness values.
2) Prescribed burns around infrastructure and sequoia groves which, while essential, do not contribute significantly to yearly burned acreage.[2]
3) Full suppression operations on fires threatening critical resources, often in lower-elevation front-country areas of the park.

While prescribed burning and full suppression tactics are well-established measures to protect values at risk in the park, managing fire for resource benefit has proven more controversial in the decades since it was introduced here in the park and elsewhere in the South Sierra Nevada (see Pyne, *Pyrocene Park*).

In addition to renewed doubts about managed fire strategies (for example Schultz, 2025, and associated media coverage), expanded availability of publicly available remote sensing data has increasingly subjected fire managers to external scrutiny (see for example (Odion and Hanson 2006, 2008; Safford, et al. 2008)). These remote sensing data streams, including higher resolution burn severity mapping using Sentinel 2 and active fire perimeter mapping using VIIRS (Chen, et al. 2022), are powerful tools for managers, ecologists, and the engaged public.

However, there are also important drawbacks and limitations to these tools, which this technical note will describe in detail for the case of the last decade of fire management in Yosemite. Inspired by similar recent analyses in the published literature (McFarland, et al. 2025), here I suggest that VIIRS daily growth perimeter analyses can complement traditional RdNBR burn severity mapping, particularly when standard burn severity analyses fail to capture

[1] Using VIIRS daily growth polygons >30 acres, I get 94582 acres at 28.4% high severity, to give a sense of the overall accuracy of the VIIRS polygon method compared to official final fire perimeters.
[2] Pile burning after thinning acres is not considered here, though it is also counted towards the yearly acreage counts.

ecological and management-relevant differences the relationship between fire behavior (growth) and fire effects (severity).

Table 1: *Management Targets for Forest Types in Yosemite NP.*

Forest Group	**Total Park Cover (Acres)**	**Target High Severity (FMH)**	**Historic High Severity (Williams, 2023)**	**Vegetation Types**
Subalpine	263098	15-35% (Lodgepole)	24% (Lodgepole)	Lodgepole Pine Forest, Whitebark pine/Mountain Hemlock Forest
Upper Montane	215970	0-15% (Red Fir)	10% (Red Fir)	Red Fir Forest, Western White Pine/Jeffrey Pine Forest, Montane Chaparral
Lower Montane	139433	5-10% (lower slopes), 30-90% (upper slopes)	5-8%	Giant Sequoia/Mixed Conifer Forest, White Fire/Mixed Conifer Forest, Ponderosa Pine/Mixed Conifer Forest, Ponderosa Pine/Bear Clover Forest, California Black Oak Woodland, Canyon Live Oak Forest
Foothill Woodlands	8752	TBD	NA	Foothill Pine/Live Oak/Chaparral Woodland, Foothill Chaparral

Methods

Fire Selection

Fire perimeters were selected and downloaded from NIFC's Historical Wildland Fires database, filtering for fires over 300 acres and within the jurisdiction of CA-YNP. Dominant vegetation type for each fire was assigned manually using the publicly available Yosemite vegetation map and visual estimates in QGIS. Ferguson fire perimeters were restricted to on/near-park fire days that affected largely Lower Montane forest cover (see Table 2).

RdNBR Mapping

RdNBR mapping was performed in Google Earth Engine (see Coding Appendix) using a dNBR mapping template from the UN SPYDER recommended practice and modified to calculate RdNBR with established Sierra Nevada severity thresholds following (Miller and Thode 2007). Fires pre-2020 were mapped using LandSat imagery with 30m resolution, and more recent fires were mapped using Sentinel with 10m resolution. Pre- and post-fire imagery selection was fire-specific based on available imagery (see Table Methods). Immediate pre and post images were selected if available, and if clouds or smoke obscured the fire area, an analogous time frame the following year was selected to minimize phenological/climatic differences between the pre/post comparison.

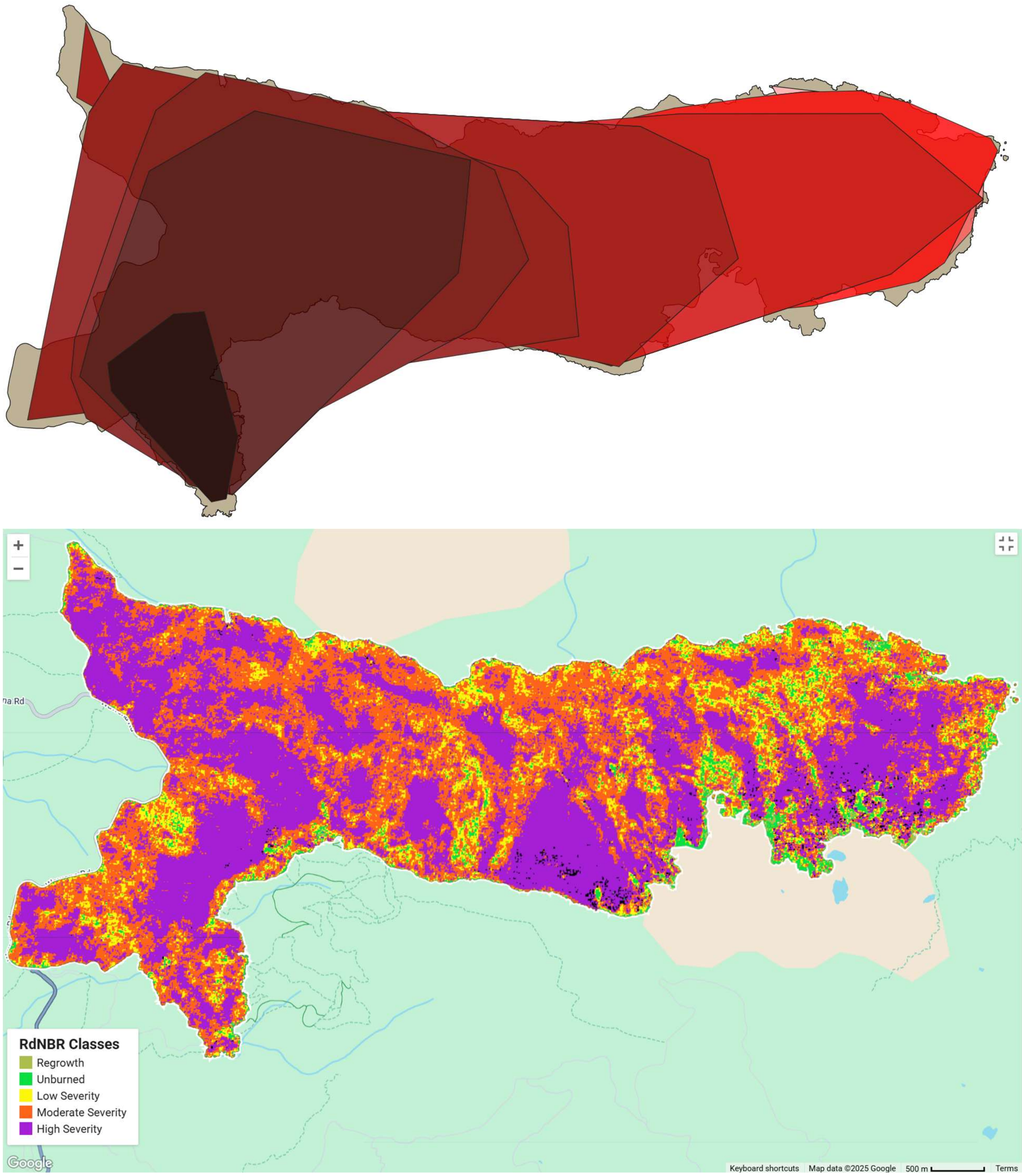
VIIRS Daily Growth Mapping

VIIRS heat detection data was requested and downloaded from NASA FIRMS for the Yosemite area from the SUOMI VIIRS C2 satellite. Active fire perimeters were then drawn following the method of (Chen, et al. 2022), though due to time constraints, we opted to use a simple convex perimeter mapping algorithm to draw polygons around new heat detections for a fire area (following Hankin et al. in the Blue Jay Fire Report) as opposed to using the more accurate but complex "alpha hull" function. Percent of high severity pixels per growth polygon was calculated in QGIS using a custom script (see Coding Appendix).

Table Methods. *Satellite image acquisition settings for RdNBR analysis*

Incident	Pre-Fire Start	Pre-Fire End	Post-Fire Start	Post-Fire End	Satellite
Blue Jay	6/15/2020	7/15/2020	7/15/2021	7/30/2021	Sentinel
Lakes	6/25/2016	7/8/2016	6/25/2017	7/15/2017	Landsat
Wolf	7/1/2020	7/15/2020	8/1/2021	8/15/2021	Sentinel
Middle-2015	7/17/2015	8/3/2015	7/17/2016	8/3/2016	Landsat
Indian	8/20/2018	9/4/2018	8/20/2019	9/4/2019	Landsat
Red	6/1/2022	6/15/2022	9/23/2022	9/30/2022	Sentinel
Rodgers	6/1/2022	6/15/2022	10/1/2022	10/15/2022	Sentinel
Tiltill	7/1/2021	7/15/2021	8/30/2021	9/15/2021	Sentinel
Lukens	6/20/2021	6/28/2021	8/15/2021	8/30/2021	Sentinel
Pika	6/22/2023	6/27/2023	8/1/2023	8/6/2023	Sentinel
Empire	7/15/2017	7/20/2017	10/27/2017	11/5/2017	Landsat
South Fork	8/1/2017	8/13/2017	8/13/2018	8/30/2018	Landsat
Tenaya	8/25/2015	9/7/2015	9/21/2015	10/5/2015	Landsat
Ferguson*	6/30/2018	7/16/2018	9/15/2018	9/30/2018	Landsat
Washburn	6/15/2022	6/20/2022	8/4/2022	8/15/2022	Sentinel

Figure Methods Example. *Example VIIRS inferred daily perimeter growth mapping (top) and RdNBR analysis (bottom) for the Washburn fire.*



Results

The majority of the acres burned in Yosemite over the past decade have not been due to intentional management decisions, but despite them; full suppression fires made up 58,076 acres with an average high severity coverage of 30.7% (Table 2). These large suppression fires included the Ferguson (2018), a fatality fire, and the Washburn (2022), which endangered the Mariposa grove of Sequoias and the town of Wawona (Hankin, et al. 2023). In contrast, fires managed under less "traditional tactics," including resource benefit, minimal impact suppression (MIST), and "mixed" strategies burned 34,506 acres with an average high severity coverage of 29.7% (Fig. 1).

To better understand the relationship between severity and fire growth, daily fire perimeters were estimated using VIIRS satellite heat detections, and daily acreage/% high severity coverage were calculated for each fire day (Fig. 2). Overall, daily fire severity did generally increase with daily growth area, but this trend did not hold true in the Subalpine vegetation group and there was only a weak correlation in the Upper Montane vegetation type. This data suggests that in these higher elevation fuel types, daily growth is primarily a function of fine fuel receptivity/immediate weather conditions while the burned area's ultimate severity is controlled by fuel loading, dominated by heavy dead and down fuels. In contrast, Lower Montane fires were characterized by a greater degree of linkage between daily growth area and severity, especially at daily growth rates of 1000 acres or more. However, the rough nature of the perimeter mapping method used (see Fig. Methods as an example) may also significantly contribute to the generally low R^2 values, so these results should be interpreted with caution.

Table 2: *Yosemite Fire Summaries (2015-2023).*

Incident	Start Date	Total Acres	% High Severity	Management Strategy	Dominant Forest Type	Forest Group
Blue Jay	7/24/2020	6895.72	22.1%	MANAGED	Red Fir	Upper Montane
South Fork	8/13/2017	7560.08	11.6%	SUPRESSION	Ponderosa - White Fir	Lower Montane
Lakes	7/8/2016	1000.72	0.8%	MANAGED	Lodgepole	Subalpine
Empire	8/1/2017	8577.55	27.5%	MIXED	Red Fir	Upper Montane
Wolf	8/11/2020	1970.29	11.1%	MANAGED	Lodgepole	Subalpine
Tenaya	9/7/2015	408.91	32.6%	SUPRESSION	Red Fir	Upper Montane
Middle	8/3/2015	324.76	1.7%	MANAGED	Red Fir	Upper Montane
Ferguson*	7/13/2018	45221	32.30%	SUPRESSION	Canyon Live Oak - Ponderosa	Lower Montane
Indian	9/4/2018	358.38	2.8%	MANAGED	Red Fir	Upper Montane
Rodgers	8/8/2022	2839.77	29.8%	MIST	Red Fir	Upper Montane
Tiltill	7/31/2021	2348.52	40.7%	MIST	Red Fir	Upper Montane
Lukens	6/28/2021	918.17	15.8%	MIST	Red Fir	Upper Montane
Red	8/4/2022	8432.39	36.8%	MANAGED	Red Fir	Upper Montane
Washburn	7/7/2022	4886.1	46.3%	SUPRESSION	Ponderosa - White Fir	Lower Montane
Pika	6/29/2023	840.23	23.2%	MIST	Red Fir	Upper Montane

**While the Ferguson burned large swathes of land over several thousand feet of elevation, both in the foothills below the park and into park, this analysis only considers the on-park parts of the Ferguson scar. These areas burned between roughly 7/29/2018 and 8/13/2018, largely in lower montane forest.*

Figure 1. *Total Acreage x Percent High Severity.*

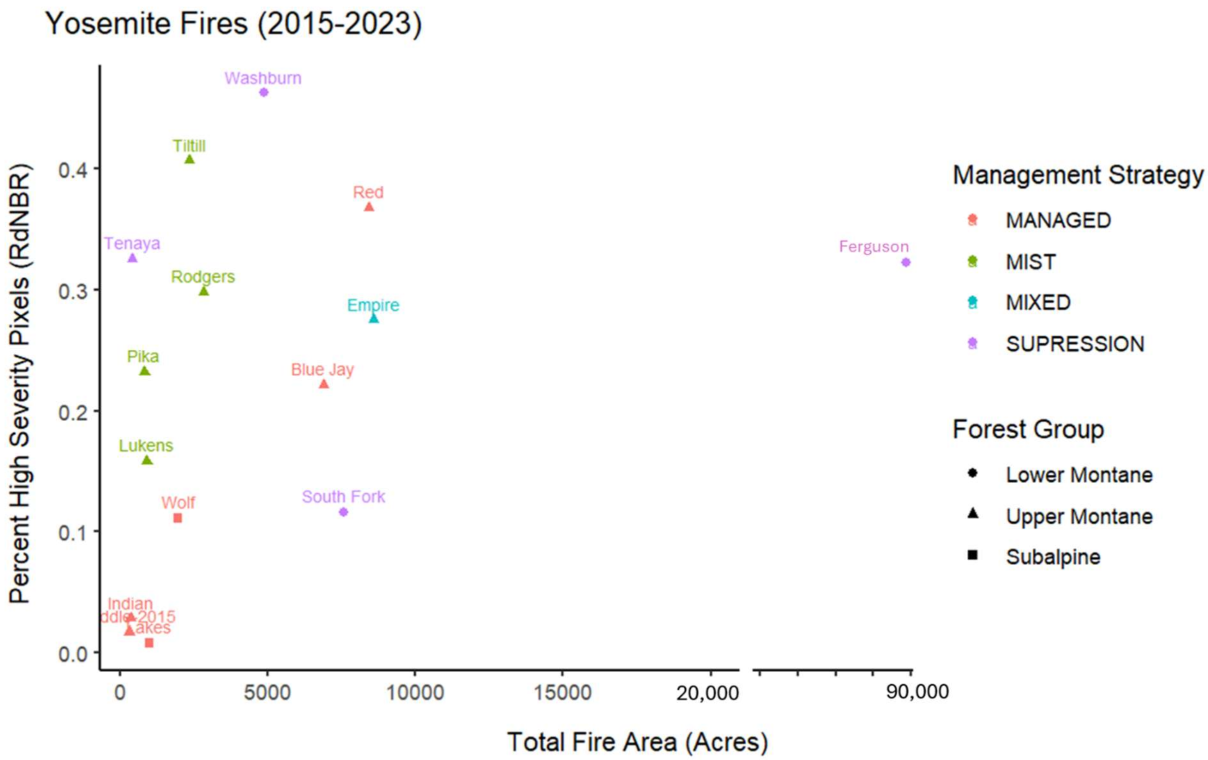
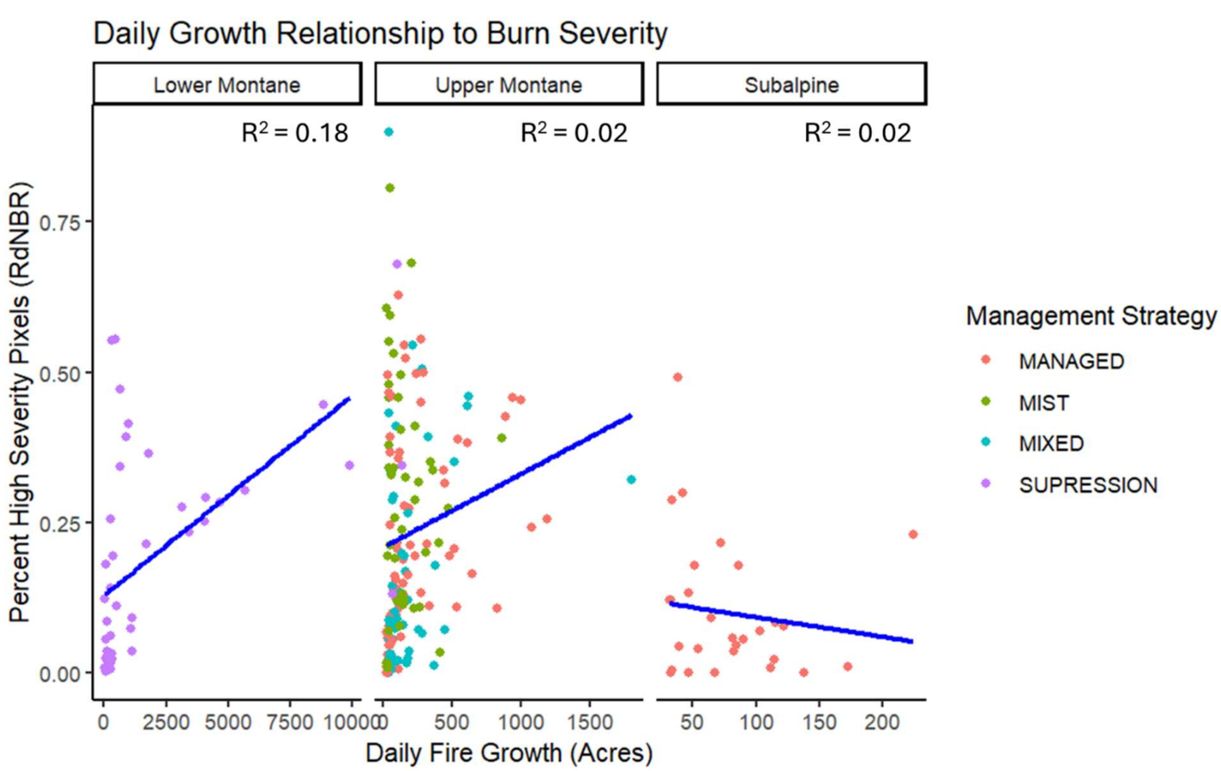


Figure 2. *Daily Acres x Percent High Severity Regressions, by Forest Type.*



Discussion

Based solely on the aggregate RdNBR analyses, one could make the case that managed fires in Yosemite are ineffective since they have a similar high severity impact footprint as their "dangerous" full suppression counterparts, both of which are well-above historic levels in both the lower and upper montane forest groups (Table 1). Leaving aside non-ecological considerations such as threats to life and property, one could alternatively use the same data to argue that we should let full-suppression fires proceed unimpeded, since their impact on the forest is similar to managed fires which have produced more resilient forests and whose higher-than-historic severity may be a transient effect of first-entry burns thinning out the forest to an appropriate tree density (see for example, this essay by Chad Hanson).

Integrating VIIRS data helps disentangle this ambiguity. Even the rough approach presented here shows that full-suppression fires in the park's lower montane forests have a distinct (more strongly coupled) relationship between growth and severity compared to their upper montane counterparts, a relationship not immediately evident if relying only on conventional RdNBR.

In the lower montane, more directly coupled fire growth and severity, especially in the larger spread rates, illustrate the management challenge Yosemite faces in re-introducing high frequency, low-severity fire into these forest types. Large runs of high intensity fire growth leading to large swathes of high severity fire effects may cause severe ecological impairment and long-term type conversion, even in fire-adapted forests (McFarland, et al. 2025). Thus, scaling the ecological effects of successful burn unit projects such as those around the town of Wawona ("Studhorse RX") will prove challenging, though greater variation in burn severity on the lower-end of the growth spectrum (less than 2000 acres, Fig. 2) suggests there may be operational space ecologically for more aggressive acreage targets than the current average burn unit size (usually dozens to hundreds of acres).

In the upper montane, there is an outstanding question of whether the observed higher-than-historical average burn severities are ecologically desirable, and future research in the park could address the threshold at which an increase in high severity associated with a first entry fire outstrips its positive ecological benefits. For example, the Pika fire burned twice the historic high severity coverage typical of a red fir forest (23% vs. 10-15%, Table 1 and 2), yet it successfully achieved its objective of lowering tree density to target maintenance conditions (see FX Crew 2024 Pika Report). In the subalpine zone, managed fires are generally more mild than the high severity conflagrations that historically defined the fire regime for these forests, though this result should be taken with caution given the long FRIDs for these forest types and relatively short time-span evaluated in this study. In these higher elevation forest groups, the overall poor linkage between growth and severity suggests that the current management paradigm has not entered a regime where a manager's ability to increase fire acreage is meaningfully constrained by a concomitant increase in fire severity.

Ultimately, despite their superficially similar RdNBR profiles, fires like the Red (managed, Red fir, PHS: 36.8%) and the Ferguson (suppression, Ponderosa, PHS: 32.3%) were very different fires requiring a different management strategy to try to achieve the desired distribution of high severity fire effects on Yosemite's landscape. Building off field-based observations and new sources of satellite information such as VIIRS will continue to be critical complements to standard satellite-based burn severity mapping for evaluating the strategies and tactics employed by Yosemite's fire management program.

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